



UNIVERSITÉ GRENOBLE ALPES / ENSIMAG

Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

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1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of..... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in

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The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes problems in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

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The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are often represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These

score vectors also often contain missing or noisy information. To handle these challenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much more efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently existing methods have in common, for example the absence of rigorous distribution-free uncertainty quantification. . In the remaining part of the section, 2.3 and ?? we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [1], on which we will build our work.

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2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method to the traditional antibiotics [3].

One of the key characteristics of of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [4]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). Similarly in environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [4]. And it also improves our understanding of viral diversity and evolution.

Host range has its benefits, but at the same time it also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which

is the cause of the infection. Therefore, we want to be able to identify the correct phage from the thousands of available ones and for that we need a fast reliable host prediction method. Traditionally, the method of determining the host range used to depend on laboratory experiments in which candidate phages were tested against a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Especially with the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interactions has increased astronomically, and this leads us to the need of having a computational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to experimental validation [5] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $X \in \mathcal{X}$ be a high dimensional feature space representing the phage genomes and let $\mathcal{H} = \{h_1, \dots, h_M\}$ be the finite set of possible bacterial hosts. In a single-host classification setting, the underlying data-generating process yields random pairs $(X, H) \sim \mathbb{P}_{X,H}$. In a general multi-host setting, the target is a non-empty subset of hosts $H^* \subseteq \mathcal{H}$, i.e., $H^* \in \mathcal{P}(\mathcal{H}) \setminus \{\emptyset\}$, where $\mathcal{P}(\mathcal{H})$ denotes the power set of $\mathcal{H}$. Our primary objective is to construct a set-valued prediction mapping:

$$C : \mathcal{X} \longrightarrow \mathcal{P}(\mathcal{H})$$

which maps a test phage $X_{n+1} \in \mathcal{X}$ to a prediction set $C(X_{n+1}) \subseteq \mathcal{H}$ that guarantees target coverage with a user-specified statistical confidence $1 - \alpha \in (0, 1)$.

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the methods have significantly evolved during this time. Prediction methods have progressed from simple rule based alignment to deep representation learning. Here, we will summarize each family of methods and their underlying mathematical assumptions along with the common limitations that motivate us for this work.

2.2.1 Sequence Similarity Approaches

The earliest host prediction tools relied on sequence alignment between a phage and bacterial genomes, assuming that similarity between the two indicates infection compatibility. This was motivated by the known transfer of genetic material between the phages and their hosts through horizontal gene transfer.

One of the tools, HostPhinder [6], works by building a reference database of phage genomes

with the known hosts. Then for a new phage in question, it assigns the host that is the host of its nearest neighbour under a k-mer similarity metric. Mathematically, let $v(g) \in \mathbb{R}^{4^k}$ be the normalised k-mer frequency vector of a genome g . So given a new test phage g_{test} and a reference database $\mathcal{D}_{\text{ref}} = \{(g_i, y_i)\}_{i=1}^N$ its predicted host is:

$$\hat{y} = \operatorname{argmax}_{y \in \mathcal{Y}} \left\{ \max_{g_i: y_i = y} \operatorname{sim}(\mathbf{v}(g_{\text{test}}), \mathbf{v}(g_i)) \right\}$$

where $\operatorname{sim}(\cdot, \cdot)$ is a similarity measure.

Another method, WISH [7], uses a probabilistic approach by modelling the genome of each candidate host, $h \in \mathcal{H}$, as a variable order Markov chain with parameter vector, θ_h . It then computes the likelihood of the new phage genome sequence, $S = (a_1, \dots, a_L)$ under each host model:

$$\ell(S \mid \theta_h) = \sum_{i=k+1}^L \log \mathbb{P}(a_i \mid a_{i-k}, \dots, a_{i-1}; \theta_h)$$

The predicted host is $\hat{y} = \operatorname{argmax}_{h \in \mathcal{H}} \ell(S \mid \theta_h)$, which maximizes this likelihood.

Despite the fact that these methods are relatively simpler, they perform reasonably well when they have a phage reference that is closely related. But for a new phage that is not very common and is only a distant relative of the phages already there in the database, they degrade sharply. And this scenario is very common in metagenomic settings [5].

2.2.2 Protein Content Approaches

Instead of comparing and aligning the whole genomes, like the methods in 2.2.1 these methods break down the phage genome into its constituent proteins and then analyze their specific features from their functional or sequence properties. This approach is motivated by the fact that the proteins responsible for infecting a host are much more evolutionary conserved than the rest of the genome [8].

RaFAH [8] (Random Forest with protein Amino acid compositions for phage Host prediction) works by clustering all the known phage proteins into protein clusters using MMseqs2 [9]. It computes the amino acid composition vector of each cluster's representative sequence and then trains a random forest classifier to predict the host genus from the protein cluster membership profile of the new phage. Let $x \in \{0, 1\}^M$ be the binary indicator that encodes which of the M protein clusters are present in a given phage. Then the random forest maps this vector to the probability simplex over the host categories $\Delta^{|\mathcal{H}|-1}$:

$$f_{\text{RF}} : \{0, 1\}^M \longrightarrow \Delta^{|\mathcal{H}|-1}, \quad \text{where } \hat{\mathbf{p}} = f_{\text{RF}}(\mathbf{x}), \quad \sum_{y \in \mathcal{H}} \hat{p}_y = 1$$

RaFAH is also relevant to this report, since we also work with similar protein based cluster

representation and the contamination problem discussed in [also arises exactly from this](#) protein-cluster construction.

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These techniques are more robust to sequence divergence than the sequence alignment methods [2.2.1](#), but they rely on the availability of explicit functional annotations, which are missing for a large percentage of the already discovered phage genomes.

2.2.3 Protein Language Models and Embedding Based Approaches

The latest host prediction methods use protein language models (pLMs) [[10](#), [11](#)]. Similar to how large language models (LLMs) are trained on massive text datasets, pLMs are neural networks trained on millions of protein sequences through self-supervised learning.

A protein sequence of length n can be represented as $A = (a_1, \dots, a_n) \in \mathcal{A}^n$, where $\mathcal{A}$ represents the standard amino acid alphabet of size 20. A is mapped by an encoder Φ_{pLM} (e.g., ProtT5-XL-UniRef50) [[10](#)] to a embedding matrix:

$$Z = \Phi_{\text{pLM}}(A) \in \mathbb{R}^{n \times d}$$

where d is the embedding dimension (typically 1024 or larger). This is followed by mean pooling along the sequence length which provides us with a fixed length embedding $z \in \mathbb{R}^d$.

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_{i,:}$$

This embedding vector, z , captures the sequence’s biochemical, structural, and evolutionary properties without requiring an explicit sequence alignment. A primary foundational model in this domain is ProtTrans/ProtT5 [[10](#)], which is trained on hundreds of millions of UniRef protein sequences.

Several tools for phage protein annotation and host prediction have been developed using these embeddings. For instance, EMPATHI [[12](#)] uses a heirarchical classification pipeline to assigns functioal labels to phage proteins. It embeds the sequences using ProtT5 and performs nearest neighbour retrieval within the embedding space, mapping them to a biological function ontology. SUBLYME [[13](#)] then extends this methodology for metagenomics which enables lysin discovery directly from environmental samples. These tools are directly involved in this work as the score matrix that is used as an input to this conformal framework is generated by a pipeline which is built on these annotations.

For host prediction, we can see in [[14](#)] that protein embeddings actually improve the predictions for the phage host interactions compared to the sequence-similarity baselines [2.2.1](#). In [[15](#)] Du et al. show that through some key protein embeddings LLMs can acheive high resolution host assignments. Such results support the theory that embedding

representations contain information that is sufficient for an accurate host-prediction. But at the same time, all these methods output point predictions, a single score or a single host label, and they do not provide us with any measure of uncertainty.

2.2.4 Limitations of Existing Methods and Need for Uncertainty Quantification

Despite the diversity and the empirical success of all the methods we have seen above, they all share a fundamental limitation, they all produce point predictions (a single predicted host or a single score) without quantifying the uncertainty of these predictions. For example, a score of $\hat{s}(s, y) = 0.87$ does not mean that there is a 87% chance of infection, it is just a relative ranking metric not a true probability of infection. More importantly none of these methods provide any mathematical guarantee for coverage (that the true host would be included in the prediction set) with a given confidence level. This lack of uncertainty quantification can be a critical problem in some scenarios, like phage therapy, where picking the wrong phage for treatment can harm the patient.

Similarly, some phages are just genuinely difficult to classify which could be due to their host range having multiple hosts or simply the protein content may be insufficient to determine the exact host with confidence. In such cases a good method should be able to return all the plausible hosts rather than just forcing a single potentially wrong host. The existing methods do not have a formal way of doing this.

There are several existing frameworks for uncertainty quantification in machine learning, but most of them are not very well suited to this problem. For example, Bayesian approaches place a prior distribution over the model parameters and then compute a posterior given the observed data. Even though these approaches are supported theoretically, the issue with is that they require strong distributional assumptions that are difficult to justify for biological score datasets. Also, computing or approximating the posterior is computationally expensive when we have a phage database with several hundred thousand genomes. Another problem is that the resulting uncertainty estimates depend heavily on the choice of the prior which introduces a degree of subjectivity which should be avoided in a clinical setup where the guarantee must be transparent and verifiable.

Other techniques, like the calibration methods include techniques like Platt scaling or temperature scaling. They adjust the raw output scores of the model so that they reflect the empirical probabilities better

$$\mathbb{P}\left(Y = y \mid \hat{s}(X, y) = p\right) = p \quad \forall p \in [0, 1]$$

For example, they make sure that among all the predictions which are assigned the score of 0.8, around 80% should be correct. This is easy to apply on top of any existing models

and is also computationally cheap, but they do not provide us with any formal finite sample guarantees. The calibration is verified empirically on a validation set but it can degrade on new data with no bounds on how far the calibrated probabilities may deviate from the truth on the unseen test set.

These observations lead us to use the conformal prediction, [16], which offers exactly the properties required here. It is distribution-free, it requires no assumption on the underlying score distribution. It provides a finite-sample coverage guarantee rather than an asymptotic or empirical one. Its only requirement is exchangeability between calibration and test data. Also, it is computationally light, it requires only the computation of empirical quantiles rather than any iterative optimisation or sampling. Instead of replacing the currently existing methods, conformal prediction complements them by taking their raw scores inputs and then provides prediction sets with proven mathematical guarantees that can be proven and are detailed in . For these reasons, conformal prediction is the framework adopted in this report.

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2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [16] and then later made accessible by Angelopoulos and Bates [17], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangeability condition. In this section we will develop the framework of conformal prediction and see

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Definition 2.1 (Exchangeability).

A finite sequence of random variables $Z_1, Z_2, \dots, Z_m$ taking values in a space $\mathcal{Z}$ is exchangeable if, for every permutation σ of the indices $\{1, 2, \dots, m\}$, the joint distribution of the permuted sequence is identical to that of the original sequence:

$$(Z_1, Z_2, \dots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, Z_{\sigma(2)}, \dots, Z_{\sigma(m)})$$

where $\stackrel{d}{=}$ denotes equality in distribution.

Or, equivalently:

$$\mathbb{P}(Z_1 \in A_1, Z_2 \in A_2, \dots, Z_m \in A_m) = \mathbb{P}(Z_{\sigma(1)} \in A_1, Z_{\sigma(2)} \in A_2, \dots, Z_{\sigma(m)} \in A_m)$$

for all measurable sets $A_1, A_2, \dots, A_m \subseteq \mathcal{Z}$ and all permutations $\sigma \in S_m$ (where S_m is the symmetric group on m elements).

Independent and identically distributed (i.i.d.) sequences are always exchangeable, but

the converse does not always hold. Thus, exchangeability is considered to be a weaker condition than i.i.d., since it requires that the observations have no special ordering. This condition can easily be met by splitting any dataset. [1, 16]

2.3.1 Mathematical Setup and Prediction Set

Let $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ be a random input-label pair, where $\mathcal{X}$ is the feature space that represents our object of interest (in our case it is the phage protein level score vectors) and $\mathcal{Y} = \{1, \dots, K\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, of n exchangeable data points drawn from an unknown joint distribution, $P_{X,Y}$. Then, for a new unlabelled test input, X_{n+1} our aim is to construct a prediction set function $C : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$ depending on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathcal{Y}$, such that the true label falls in in the set with a defined coverage that is:

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ is the significance level. In our experiments we set $\alpha = 0.1$ so that the guarantee is 90%.

A prediction set function can provide us three qualitatively different outcomes for any new test point:

- (i) $|C(x)| = 0$ (Abstension): this means that the model cannot find any plausible host. This happens when the score profile of the text phage is very different from anything seen while calibrating.
- (ii) $|C(x)| = 1$ (Confident Prediction): this means that exactly one host label is consistent with the calibration distribution at $1 - \alpha$ significance level.
- (iii) $|C(x)| \geq 2$ (Unsure): this means that according to the model there are several plausible hosts.

The expected set size, $\mathbb{E}[|C(x)|]$ can help us determine the efficiency of the model. If the coverage is maintained, then the smaller the average set size the more efficient the model is, as it provides us with a more informative prediction. An ideal predictor should achieve a coverage $\geq 1 - \alpha$ with the smallest possible set size. This is the goal that we will work on in .

2.3.2 Split conformal prediction

In this work, we have used split conformal prediction, in this the training data is split into two sets: a training set and a calibration set.

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The first set, the training set, is used to fit a score model $\hat{f} : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$. We then define a nonconformity score function: $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ based on $\hat{f}$. This score function is used to calculate the error margins, it quantifies how unusual the pair (x, y) is relative to the training data.

Then the calibration set (the second set), $\{(X_i, Y_i)\}_{i=1}^n$, which is separate from the training set, is used to evaluate the nonconformity scores:

$$S_i = S(X_i, Y_i), \quad \forall i = 1, \dots, n$$

To guarantee that our error rate is less than α we need to compute the conformal calibration quantile, $\hat{q}$, from these calibration scores:

Definition 2.2 (Conformal Quantile).

The conformal calibration quantile q_α is defined as the empirical $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ -th quantile of the sorted calibration scores $\{S_1, \dots, S_n\}$:

$$q_\alpha := \text{Quantile} \left(\{S_1, \dots, S_n\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right)$$

Alternatively, using the empirical distribution function:

$$q_\alpha = \inf \left\{ q \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbf{1}(S_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\}.$$

Then, for a new input, X_{n+1} the prediction set is constructed by collecting all the candidate labels in $y \in \mathcal{Y}$ whose nonconformity scores do not exceed this calibrated threshold:

$$C(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid s(X_{n+1}, y) \leq q_\alpha \right\} \quad (2.3.1)$$

By the exchangeability assumption 2.1, we get that the test score, $S_{n+1} = S(X_{n+1}, Y_{n+1})$ has rank which is uniformly distributed among $\{S_1, \dots, S_n, S_{n+1}\}$. We get the factor of $\frac{n+1}{n}$ due to the finite-sample exchangeability assumption of the n calibration points added to the 1 unseen test point, the probability is split across $n+1$ equal segments. Multiplying $(1-\alpha)$ by $\frac{n+1}{n}$ guarantees exact finite-sample coverage [18]:

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) = \mathbb{P}(S_{n+1} \leq q_\alpha) \geq 1 - \alpha.$$

Theorem 2.3 (Marginal Coverage Guarantee [16]). *If the data points $Z_1, \dots, Z_{n+1}$ are exchangeable, then the prediction set $C(X_{n+1})$ defined in Equation (2.3.1) satisfies:*

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

Furthermore, if the nonconformity scores V_i are almost surely distinct (continuous random variables), the coverage bound is tight:

$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in C(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}$$

This mathematical guarantee in Theorem 2.3 holds exactly in finite samples without any assumptions on the nonconformity score function, s , or on the distribution of (X, Y) . The choice of S affects the efficiency as it impacts how tight the prediction sets are, but it does not effect the validity of the method. This makes separation makes conformal prediction powerful, as it enables us to use any score function and yet have the coverage guarantee [1].

Overall, this is a valid technique that achieves the coverage guarantee over the calibration set in a single pass which makes it computationally efficient.

2.3.3 Marginal & Conditional Coverage

Standard conformal prediction satisfies marginal coverage, which means that the $(1 - \alpha)\%$ coverage guarantee holds on average across all possible choices of the calibration set, D_{cal} , and test samples, (X_{n+1}, Y_{n+1}) :

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

We can also express this as an expectation over both D_{cal} and the test pair as:

$$\mathbb{E}_{D_{\text{cal}}, (X_{n+1}, Y_{n+1})} [\mathbb{I}(Y_{n+1} \in C(X_{n+1}))] \geq 1 - \alpha$$

Even though the marginal coverage is a powerful guarantee but it does not ensure the conditional coverage, $\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha$ for all x . Also, for all different classes, y it cannot ensure class conditional coverage: $\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid Y_{n+1} = y) \geq 1 - \alpha$. So, for a dataset that has class imbalance, which is very common in phage-host interaction, a model that only gives is a marginal coverage guarantee can potentially under-cover a rare small class while over cover the majority class and still maintain the average coverage [1, 19].

For example, suppose we have $\dagger = \{-1, 1\}$, with $\mathbb{P}(Y = 1) = 0.1$ and $\mathbb{P}(Y = -1) = 0.9$. Then a predictor that achieves 100% coverage for $Y = -1$ and 0% for $Y = 1$ will still manage to satisfy the marginal coverage of 90%. So, in a phage-hot classification setting, where there is asymmetry in the calibration set sizes for different hosts, a predictor may undercover the minority class below the $1 - \alpha$ target coverage.

This limitation motivates the class-conditional calibration coverage structure in . In this framework, originally introduced under Mondrian conformal prediction [16], the guarantee

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hold for each class individually:

$$\mathbb{P}(Y \in C(X) \mid Y = y) \geq 1 - \alpha \quad \forall y \in \mathcal{Y}$$

It is impossible to get the feature-conditional coverage ($X = x$) without additional assumptions, as it would require the conditional coverage at every point x , which is impossible with finite data [1]. However, we can get the class-conditional coverage ($Y = y$) is exact and valid for finite-sample. We can compute a separate threshold for each class using only the calibration points with the true label y . This is like class conditional calibration, which is a justified approximation .

This technique has been used in this report and is formalised in . The main point here is that this class conditional calibration still uses the exact same quantile argument as the marginal calibration, it just applies it separately to each class. Therefore, it is mathematically valid under the assumption of exchangeability within each class.

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2.4 Conformal Set Aggregation (CSA)

The standard split conformal framework in 2.3.2, takes a scalar non-conformity score $S(x, y) \in \mathbb{R}$. In this report, the prediction pipeline produces a score vector of scores $s \in [0, 1]^K$ for each phage, where K is the number of functions. We may lose the geometric structure of the score distribution if we aggregate this vector to a scalar value before applying conformal prediction. For such settings, Ochoa Rivera, Pater and Tewari [1] have proposed a score based aggregation (CSA) method for conformal prediction.

2.4.1 Ensemble Score Vectors

Let $\mathcal{S} \subseteq [0, 1]^K$ be the score space and let $s = (s_1, \dots, s_K) \in \mathcal{S}$ be the vector of scores produced by the K models, for each phage. Now, the CSA method treats the entire vector as the object of inference rather than reducing it into a single value. The main idea is to define a region, called an envelope, $E \subseteq [0, 1]^K$. This envelope consists of the region where the score vectors of the true hosts for that particular phage live.

Mathematically, we can say that we apply a label dependent transformation $\phi_y : [0, 1]^K \rightarrow [0, 1]^K$ to the score vector, before actually testing if it is part of the envelope. This transformation needs to be such that for a phage with the true label y , ϕ_y should be close to the origin in the score space. So, we can write our prediction set as:

$$C(s) = \{y \in \mathcal{Y} : \phi_y(s) \in \hat{t}\hat{E}\},$$

where $\hat{t}$ is the global scaling factor that is calibrated on the calibration set 2.3.2. The ϕ_y chosen in this work is defined in .

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2.4.2 The CSA Framework

The core innovation of the CSA method is in the calibration process. It splits the calibration data into two subsets which have different roles. This is necessary for the exchangeability argument that provides us with the coverage guarantee. If we used the same calibration points to construct both, we would get overfitting which would introduce bias which would invalidate the distribution free coverage guarantee [1].

The first set is used to determine the geometric shape of the optimized envelope. Let's call it the 'Shape Discover Set', it is used to map the distribution of the scores and then learn the geometry of the multi-dimensional envelope $E \subseteq \mathbb{R}^K$. This envelope is such that the transformed score vectors of the true positives are concentrated here. This set is only used to fit the shape of the conformal region without fixing the size.

The second set or the 'Size Scaling Set' is used to determine the global scaling factor $\hat{t} \in \mathbb{R}^+$. This scaling factor either inflates or deflates the envelope, E learned from the shape discovery set, depending on the value of $\hat{t}$, so that we get the precise finite sample coverage guarantee. This $\hat{t}$ is the smallest scaling factor such that the scaled envelope $\hat{t}E$ achieves the required $1 - \alpha$ coverage. The scaled envelope is defined linearly as:

$$\hat{t} \cdot E = \{t \cdot x : x \in E\}, \quad \hat{t} > 0$$

Finally, for a test input a candidate label, y is included in the prediction set its transformed score vector falls inside the scaled envelope:

$$C(s) = \{y \in \mathcal{Y} : \phi_y(s) \in \hat{t} \cdot E\}$$

where ϕ_y is a maps the raw scores into the score space.

2.4.3 Limitations of CSA in Classification

This CSA framework was originally developed and analyzed in a regression setting [1].

While effective for multi-target regression, the original CSA formulation by Ochoa Rivera et al. has major limitations when applied to discrete classification problems: - Assumption of Convexity: The original paper assumes that E is a convex hull or intersection of half-spaces. In bounded classification settings, the score vectors for distinct

classes often concentrate in separate clusters near different boundaries of the $[0, 1]^K$ hypercube. Forcing a convex envelope across these clusters creates a region that covers large areas of empty score space between them. This significantly inflates the volume of the envelope, leading to overly conservative, uninformative, and large prediction sets $|C(\mathbf{s})|$.

- Iterative Calibration Inefficiency: The original method relies on iterative numerical optimization or binary search to find the optimal scaling factor $\hat{t}$ over S_2 . For high-dimensional score spaces or massive metagenomic datasets, this iterative approach is computationally expensive and lacks mathematical transparency.
- Lack of Per-Class Guarantees: The original scaling calculation enforces only a marginal coverage guarantee, which can lead to poor predictive performance on highly imbalanced biological datasets. These limitations motivate the core contributions of this thesis: designing structured non-convex envelopes, developing an exact analytical solution for $\hat{t}$, and implementing class-conditional adjustments to ensure robust statistical guarantees for phage-host range prediction.

3 Methods

3.1 Problem Formalization

- Input: a phage represented by a score vector in $[0, 1]^K$ where K is the number of functional dimensions
- Define the label space $\mathcal{Y}$ (finite set of classes)
- Goal: construct a prediction set $C(x) \subseteq Y$ with the conformal guarantee
- The one-envelope design: build one single envelope on all training phages under their true-label NC transformation, rather than separate envelopes per class

3.2 Non Convex envelope for classification

- In the regression setting of the original CSA paper, envelopes are typically convex because the score space is unbounded and the data does not cluster into separated groups
- In our classification setting, the score space is bounded in $[0, 1]^K$ and — crucially — true NC vectors for different classes occupy distinct, well-separated regions of this space (example)
- In the classification, produce tighter prediction sets without sacrificing the coverage guarantee
- The guarantee still holds because it depends only on the exchangeability of the data and the quantile construction, not on the shape of the envelope

3.3 Score Transformation

- General formulation: for each class $y \in \mathcal{Y}$, define a transformation $\phi_y: [0, 1]^K \rightarrow [0, 1]^K$ that maps the raw score vector to an conformal score vector such that a true infector of class y is near the origin
- Property we require: $\phi_{i_y}(\mathbf{s})$ is small when $\mathbf{s}$ is consistent with label y , large otherwise
- we make a single unified envelope coherent across all classes

3.4 Envelope Constriction

3.4.1 CSA Two-Stage Calibration

- Training vectors are split into two disjoint halves S_1 and S_2 (50/50 random split).
- S_1 (shape discovery): used to learn the geometry of the envelope.
- S_2 (size scaling): used to calibrate the scaling factor $\hat{t}$ that determines the final size of the envelope.

- The two stages must use disjoint data to preserve the exchangeability argument underlying the coverage guarantee.

3.4.2 τ Computation

- Define $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$, the minimum scaling factor needed to include point $\mathbf{s}$ in the scaled envelope E .
- The global scaling factor is then determined by the $(1 - \alpha)$ quantile of the τ values over S_2 :

$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1 - \alpha) \rceil$$

where $\tau_{(k)}$ denotes the k -th smallest value of τ in S_2 .

- Key contribution: for each of the three envelope geometries, $\tau(\mathbf{s})$ has a closed-form expression computable in a single pass — no iterative search required.
- Computational benefit: the entire calibration step reduces to one forward pass over S_2 followed by a sort.
- Contrast with the original CSA paper which uses iterative binary search.

3.4.3 Radial Method

3.4.4 Strip Method

3.4.5 1D

3.4.6 Class-Specific $\hat{t}$ Calculation

3.4.1 — CSA Two-Stage Calibration

3.5 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data

4.1 Data Description

- The two files: protein-level score matrix (1,853,074 proteins $\times$ 40 functional score columns, values in $[0,1]$, NaN when protein not annotated with that function) and meta-data file (same 1,853,074 rows, columns: accession, host, host_type, split)
- Scale of the dataset: 18,474 unique phages after aggregation
- The split column: integer 0–4 encoding a genome-cluster-based partition — phages in the same genome cluster are always in the same fold, ensuring test phages are genuinely different from training phages at the sequence level
- Class distribution: gram-neg vs gram-pos vs unknown
- The NaN structure: NaN is structural missingness, not noise — a protein that was not annotated with a given function simply has no score for that function, which is different from a score of zero

4.2 Protein to Phage Aggregation

- The raw data is at the protein level but conformal prediction operates at the phage level — aggregation is needed
- NaN-aware mean pooling: for each phage and each functional score column, compute the mean across all proteins of that phage that have a non-NaN value for that column. Proteins with NaN for a given column are excluded from the mean for that column only
- Why not replace NaN with zero before pooling: treating an unannotated protein as having zero evidence would push the phage’s aggregated score artificially toward zero, misrepresenting its biology
- After aggregation: result is a matrix of shape (number of phages) $\times$ 40, still with NaNs in cells where none of the phage’s proteins were annotated with that function

4.2.1 NaN handling after aggregation

- After mean pooling, some phages still have NaN in certain columns -> this happens when none of their proteins were annotated with a particular function
- Strategy: fill remaining NaNs using column-wise medians computed strictly on the training set (never the test set, to prevent data leakage) - The median cap: any training median above 0.6 is capped at 0.5 -> this prevents an imputed value from falsely pushing a phage toward a particular class - any remaining NaNs after median imputation are filled with 0.5 (neutral value)

4.3 Train-Test Split

- Standard split: split 0 is the held-out test set, splits 1–4 form the training set
- The contamination problem: explain the protein cluster (pc) column -> a protein's cluster may include proteins from phages in different folds. The models that produced the scores were trained on protein clusters, so a protein from a training-fold phage may share a cluster with a test-fold protein, introducing subtle leakage
- The masking fix: the file `all_random_pc_splits.pkl` encodes for each protein and each (host, function) model which fold that protein was assigned to. Setting scores to NaN where the protein was not in the test set of the relevant model, before aggregation, removes the contaminated predictions. The existing NaN-aware mean pooling then handles these naturally -> this is the "missing value" framing

5 Experiments and Results

5.1 Synthetic Data Validation

5.1.1 Purpose and setup

- Why synthetic validation before real data: allows controlled verification that the coverage guarantee holds exactly, with known ground truth distributions
- General setup: simulate score vectors from known distributions, apply all three envelope methods, verify empirical coverage matches theoretical target $1 - \alpha$

5.1.2 2D Experiments

- Three scenarios tested in 2D ($K=2$): - Uncorrelated scores: two independent uniform or Gaussian dimensions
- Correlated scores: scores with positive correlation -> tests whether the envelope adapts to the correlation structure
- Asymmetric predictors: one informative dimension and one uninformative dimension -> tests whether the envelope correctly ignores the noisy dimension
- For each scenario: show the envelope shape visually, report empirical coverage across multiple runs, confirm it matches $1 - \alpha$
- Key observation from 2D: the radial and strip envelopes are visibly non-convex and wrap tightly around the data, while collapsed 1D produces a simple interval

5.1.3 Extension to K Dimensions

- Repeat the experiments for $K \in 2, 5, 10, 20$ with a single host
- Show that all three methods maintain coverage $\geq 1 - \alpha$ across all dimensionalities
- Discuss the effect of increasing K on envelope quality: radial method becomes sparser (fewer S1 points per sector), strip method has more dimension pairs to condition on, collapsed 1D is unaffected
- Show that the analytical τ computation gives identical results to what iterative search would give, confirming correctness

5.1.4 Multiple Hosts

- Extend to a simulated multi-class setting with multiple hosts ($|\mathcal{Y}| = 6, K = 5$)
- Each host's true score distribution is a distinct cluster in $[0, 1]^K$
- Verify that the single unified envelope correctly assigns the true host to the prediction set with probability $\geq 1 - \alpha$ - Show 3D visualisation of the envelope for $K=3$

5.2 Gram Type Classification

5.2.1 Experimental Setup

- Data: split-0 phages only (80% calibration, 20% test), unknown host types excluded
- Parameters: $\alpha = 0.1$, $M = 200$ directions (radial), $M=20$ bins (strip), 10 independent runs to estimate variance
- The NC transformation used: keep scores as-is for one class, flip with $1 - s$ for the other

5.2.2 Results

- table: overall coverage, per-class coverage, average set size, empty set rate, size-2 rate, singleton accuracy std across 10 runs for all three methods
- Bar charts: set size distribution, overall and per-class coverage

5.3 Host Classification

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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